

1. **Data Integrity:** This refers to the quality of the data. It is important because computers rely on data to generate information, for that information to be valuable, the underlying data must be accurate, verifiable, and reliable.

Garbage In, Garbage Out (GIGO): This phrase means that the accuracy of a computer's output depends on the accuracy of the input. If you enter incorrect data ("garbage in"), the computer will produce incorrect information ("garbage out")

2. **Validation:** This is the process of comparing data with a set of rules or values to determine if the data is correct.

Types of Validity Checks: The text lists the following checks:

- Alphabetic check
- Numeric check
- Range check
- Consistency check
- Completeness check
- Check digit

3. **Data Redundancy:** This is a weakness because duplicating data wastes storage space and can lead to data inconsistency

Isolated Data: This is a weakness because storing data in separate, independent files makes it difficult for different departments or programs to access and share that data.

4. **Database Approach:** This is a system where many programs and users share the data stored in a central database, rather than having separate files for each application.

Strengths:

- Reduced data redundancy
- Improved data integrity
- Shared data
- Easier access
- Reduced development time

5. **Data Dictionary:** It contains data about each file in the database and each field within those files

Query Language vs. QBE: A query language consists of simple, English-like statements that users type to specify data actions (like display, print, or store). Query by example (QBE), on the other hand, provides a graphical user interface (GUI) where users interact with visual elements (like forms or lists) to retrieve data rather than typing commands

Checkpoint

The Checkpoint exercises test your knowledge of the chapter concepts. The page number containing the answer appears in parentheses after each exercise.



To complete the Checkpoint exercises interactively, visit the Computer Concepts CourseMate Web site at www.cengagebrain.com and then navigate to the Chapter 9 Checkpoint resource for this book.

Multiple Choice

Select the best answer.

- A _____ is a field that uniquely identifies each record in a file. (356)
 - data type
 - data file
 - ☒ primary key
 - data character
- In some tables, the primary key consists of multiple fields, called a _____. (356)
 - ☒ composite key
 - secondary key
 - key field
 - column
- _____ procedures include adding records to, modifying records in, and deleting records from a file. (357)
 - Range check
 - ☒ file maintenance
 - Backup
 - Validation
- ☒ _____ verifies that a required field contains data. (360)
 - ☒ completeness check
 - range check
 - consistency check
 - numeric check
- When an organization uses _____, many programs and users share the data in the database. (361)
 - a data model
 - a file processing system
 - ☒ the database approach
 - a check digit
- All of the following are strengths of the database approach, except _____. (362)
 - ☒ less complexity
 - improved data integrity
 - reduced development time
 - easier access
- Many organizations adopt the _____ policy, where users' access privileges are limited to the lowest level necessary to perform required tasks. (367)
 - golden rule
 - ☒ principle of least privilege
 - principle of highest access
 - least order
- The database analyst (DA) _____. (371)
 - monitors the performance of the database
 - creates and maintains the data dictionary
 - ☒ decides on the proper placement of fields
 - checks backup and recovery procedures

Matching

Match the terms with their definitions.

a. numeric check

u. checks backup and recovery procedures

Matching

Match the terms with their definitions.

e 1. field name (355)

C 2. field size (355)

q 3. query language (364)

d 4. recovery utility (367)

f 5. database administrator (371)

~~X~~ consists of simple, English-like statements that allow users to specify the data to display, print, or store

b. item that contains data and the actions that read or process the data

~~X~~ defines the maximum number of characters a field can contain

~~X~~ uses the logs and/or backups to restore a database when it becomes damaged or destroyed

~~X~~ uniquely identifies each field

~~X~~ creates and maintains the data dictionary, manages security of the database, monitors the performance of the database, and checks backup and recovery procedures